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Durability of glass FRP bar-reinforced seawater–sea sand concrete beams under sustained loading in a subtropical coastal environment

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Abstract

The replacement of conventional bars with fiber-reinforced polymer (FRP) bars in seawater–sea sand concrete (SSC) structures is a promising method for sustainable construction. To promote the engineering application of the aforementioned structures, the primary objective of this study was to investigate the concerns regarding the durability of SSC beams reinforced with GFRP bars. For this purpose, three-point bending tests were conducted on GFRP bar-reinforced SSC beams to examine the coupled effects of sustained loading and exposure to a subtropical coastal environment on the flexural behavior of exposed SSC beams. The test results revealed that SSC beams reinforced with GFRP bars exhibited excellent durability in a subtropical coastal environment. The coupled effect of sustained loading and exposure to the subtropical coastal environment had an insignificant negative influence on the ultimate flexural capacity of GFRP bar-reinforced SSC beams at 540 days of age. The failure mechanism of the exposed GFRP bar-reinforced SSC beams was concrete crushing due to over-reinforcement. Finally, the flexural capacity of the exposed beams was predicted by a function of the degradation in the tensile strength of the GFRP bars.

Highlights

- The coupled effects of loading and environmental exposure on GFRP-SSC beams were examined.
- The failure mechanisms of the exposed GFRP-SSC beams was investigated.
- The long-term flexural capacity of the GFRP-SSC beams was predicted.

KEYWORDS

flexural durability, GFRP bar, seawater–sea sand concrete beams, subtropical coastal environment, sustained loading

1 | INTRODUCTION

The scarcity of natural raw materials for construction, for example, fresh water and river sand, has emerged as a concern owing to the massive consumption of concrete in

recent decades.¹ In an attempt to alleviate the shortage of materials for construction, plentiful marine resources are considered as a potential alternative for construction materials; for example, seawater and sea sand could be used to replace fresh water and river sand,

respectively.^{2–4} In coastal construction projects, the adoption of seawater and sea sand as the dominant raw materials will facilitate the production of concrete and reduce transportation costs and construction periods.^{5–7} For this reason, seawater–sea sand concrete (SSC) has been developed. Moreover, fiber-reinforced polymer (FRP) bars with excellent mechanical properties that are insensitive to corrosion by chloride ions,^{8,9} which show favorable potential for coastal engineering application due to the lack of steel bar corrosion.^{10–12} Corrosion-free glass fiber-reinforced polymer (GFRP) bars are applied in many engineering cases due to their high strength, cost efficiency and mature production technology.^{13–16} Hence, the application of GFRP bars as reinforcements for SSC in coastal structures shows promise and has attracted widespread attention.¹¹

A previous study indicated that the effect of chloride ions from seawater and sea sand on the flexural capacity of GFRP bar-reinforced SSC beams was insignificant and that the ultimate load strongly depended on the reinforcement ratio and shear-span ratio¹⁷; moreover, the ultimate load of GFRP bar-reinforced SSC beams was greater than that of steel-reinforced beams under the same reinforcement ratio.¹⁸ Furthermore, GFRP bars have the potential to be used as prestressing bars due to the evolution of materials and production processes.^{19–21} The resistance of prestressed GFRP bar-reinforced SSC beams to cracking is significantly enhanced by prestressing.¹⁷ However, engineering structures are commonly subjected to loading under service conditions, and the coupled effects of loading and the environmental exposure accelerate the degradation of structures.^{22,23} Complicated service conditions may negatively affect the anticracking and flexural behavior of GFRP bar-reinforced SSC beams.

Numerous tests have been conducted on the durability performance of nonprestressed GFRP bar-reinforced beams in recent decades. Chen et al.²⁴ investigated the ultimate flexural capacity of nonprestressed GFRP bar-reinforced SSC beams after 270 days of exposure to typical atmospheric and tidal environments. The results showed that the ultimate flexural capacity of the beams in the tidal environment decreased by 13% compared to that of the beams in the general atmospheric environment. Ren et al.²⁵ reported an insignificant degradation in the ultimate flexural capacity of normal concrete (NC) and SSC beams reinforced with nonprestressed GFRP bars exposed for 12 months in an outdoor subtropical coastal environment. In addition, Chang et al.²⁶ investigated the flexural strength of nonprestressed GFRP bar-reinforced SSC beams exposed to seawater and a sustained load (40% of the load level) for 270 days and found that the strength of the beams decreased by 40% over that

period. However, Esmaeili et al.²⁷ reported that the flexural strength of nonprestressed GFRP bar-reinforced concrete beams decreased by only 16% after the coupled effect of sustained loading (40% of the load level) and exposure to a field environment at Halifax for 10 years. The above studies suggested that the results of accelerated environmental tests may yield more conservative predictions than results obtained under real-life conditions.²⁸ Wang et al.²⁹ investigated the mechanical properties of prestressed and nonprestressed GFRP bar-reinforced beams coupled with accelerated seawater wet–dry cycling and sustained loading for 270 days. It was found that there was insignificant deterioration in the flexural behavior of the beams under these conditions, but their anticracking behavior significantly decreased due to the loss of prestressing. Notably, field durability testing of prestressed GFRP-reinforced beams has received little attention. Therefore, to determine the real service life of GFRP bar-reinforced SSC beams in detail, investigating the durability of prestressed/nonprestressed GFRP bar-reinforced SSC beams under coupled field environmental exposure and sustained loading is highly important.

The excellent resistance of GFRP bars to saltwater is quite promising for use in coastal engineering,^{30,31} but changing the environmental conditions and the alkaline environment inside the concrete may trigger the erosion of fibers, matrix hydrolysis and debonding at fiber–matrix interfaces in GFRP bars.^{32–34} Damage to GFRP bars starts mainly around their exterior surfaces, as this is where they are in contact with the concrete.³⁵ This implies that the deterioration of GFRP bars in concrete will negatively affect the bonding between GFRP bars and concrete, thus reducing the durability of GFRP bar-reinforced concrete beams. As such, the degradation of GFRP bars under real service conditions has always been a concern. Previous studies have focused on investigating the degradation of GFRP bars in accelerated aging environments.^{35–39} The mechanical properties of GFRP bars were revealed to degrade significantly in highly alkaline environments (from a pH of ~ 12.7 to 13.4).^{35,40,41} Fortunately, enhancement of the durability of GFRP bars has recently been reported. D'Antino et al.⁴² investigated the tensile performance of new thermoplastic GFRP bars produced by improved manufacturing techniques exposed to a 60° alkaline solution for 1000 h and found that the tensile strength and elastic modulus retention of thermoplastic GFRP bars were 87.6% and 91.7%, respectively. Wang et al.⁴³ investigated the evolution of the mechanical properties of GFRP bars in low-alkali concrete (pH of ~ 11 or 12) and reported that the tensile strength retention of GFRP bars in the low-alkali concrete (pH of ~ 11) increased by 17.0% compared to that in NC (pH of ~ 13). However, although accelerated aging methods can be

used to investigate the degradation mechanisms of GFRP bars, the test results may lead to conservative predictions for GFRP bars in real environments because laboratory conditions cannot replicate the synergistic effects present in real environments.

The present study investigated the effect of a subtropical coastal environment on the tensile performance of SSC-coated GFRP bars (4 control specimens and 16 conditioned specimens) and the flexural performance of SSC beams reinforced with GFRP bars (2 control specimens and 8 conditioned specimens). The exposure times were 90, 180, 270 and 540 days. A sustained load was applied to the SSC beams during exposure in the field. The relationship between the degradation in the tensile strength of GFRP bars and the flexural behavior of GFRP bar-reinforced SSC beams was examined.

2 | EXPERIMENTAL PROGRAMME

2.1 | Raw materials

The fine aggregates (sea sand) and coarse aggregates (granite gravel) used in the SSC were sourced from a supplier in Guangzhou, China. The chloride ion concentration of the sea sand was 0.02%. The mean particle size of the coarse aggregates ranged between 5 and 20 mm. Common silicate Portland cement (PO 42.5R) was adopted. According to the guidelines of the code ASTM D1141,⁴⁴ artificial seawater was prepared and used in the concrete mix. Table 1 presents the mix proportions of SSC and the compressive strength of the resulting SSC (100 mm × 100 mm × 100 mm) at different exposure times.

GFRP bars were produced by Haichuan New Material Technology Co., Ltd., Shenzhen, China. An advanced curing system was used for the GFRP bars to increase their stability. Following previous studies on SSC beams,¹⁷ GFRP bars with nominal diameters of 14 and 10 mm were employed in this paper as the tensile and compressive reinforcing bars of the test beams, respectively. Vinyl resin was used for the matrix of the GFRP bars. The surface of the GFRP bars were sand-coated. The stirrups were made from a unique continuous bar with a lap length of 130 mm and a corner radius of 30 mm. The interlaminar shear strength of the 10 mm diameter stirrups was 140 MPa.

Based on the code ASTM D7205,⁴⁵ the ultimate tensile strength and elastic modulus of the tensile GFRP bars were measured to be 1102 MPa and 51.72 GPa, respectively.

2.2 | SSC-coated GFRP bar details and tensile test

GFRP bars are embedded in concrete under actual engineering applications. To investigate the effect of exposure conditions on GFRP bars inside SSC beams, specimens of SSC-coated GFRP bars were cast and placed in the same locations as the SSC beams for environmental exposure. As shown in Figure 1A, the length of the GFRP bars coated with SSC was 800 mm, and the diameter of the GFRP bars was 14 mm. A polyvinyl chloride (PVC) pipe was adopted as the mold for casting the concrete. GFRP bars were fixed in the centre of the PVC pipe, after which SSC was poured into the pipe. The thickness of the SSC coating was 30.5 mm. The SSC-coated GFRP bars were cured for 28 days under the same conditions as those used for the SSC beams. The specimens were subjected to environmental exposure tests after they were cured.

At specific exposure times (i.e., 90, 180, 270 and 540 days), the SSC was carefully broken with a small hammer to extract the GFRP bars. In accordance with the ASTM D7205 standard,⁴⁵ tensile tests were conducted on the GFRP bars at a displacement rate of 5 mm/min. As shown in Figure 1B, a steel pipe was anchored to each end of the GFRP bars with epoxy resin, and the length of each pipe was 200 mm. The elongation of the GFRP bars was measured by an extensometer with a range of 50 mm.

2.3 | Details of the GFRP bar-reinforced SSC beams and three-point bending test

Based on previously reported results on the short-term flexural behavior of SSC beams reinforced with GFRP bars,¹⁷ as shown in Figure 2A, an SSC beam with dimensions of 150 mm × 220 mm × 1200 mm was adopted. Two 10 mm GFRP bars and four 14 mm GFRP bars were deployed in the compression and tension zones of the beam, respectively. The diameter and spacing of the stirrups were 10 and 50 mm, respectively. Table 2 lists all the specimens in detail. According to the literature,¹⁷ half

TABLE 1 Mixture design and compressive strength of SSC.

Seawater (kg/m ³)	Cement (kg/m ³)	Sea sand (kg/m ³)	Coarse aggregate (kg/m ³)	Compressive strength (MPa)				
				Control	90 d	180 d	270 d	540 d
210	456	570	1159	47.74	54.19	58.18	59.56	64.01

FIGURE 1 Test setup of the GFRP bar: (A) diagram of the SSC-coated GFRP bar (dimensions in mm) and (B) tensile test of the GFRP bar.

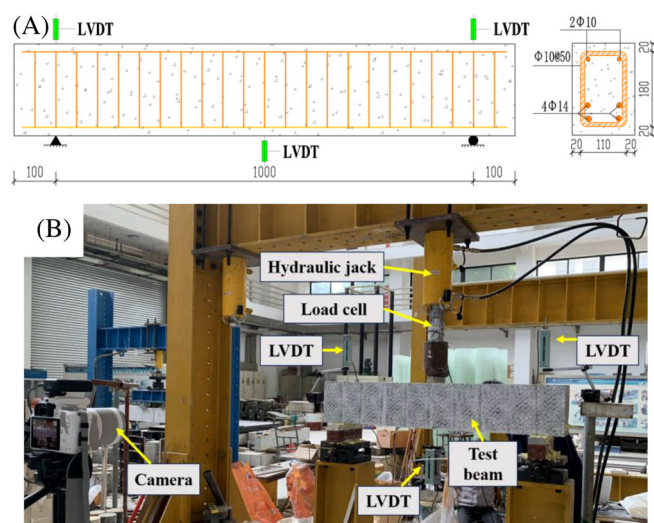
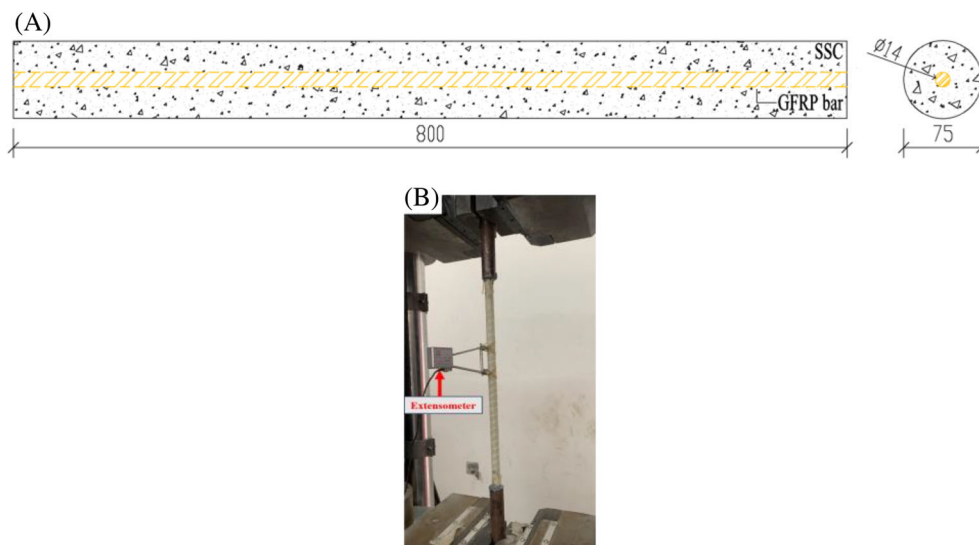


FIGURE 2 Test setup for an SSC beam: (A) diagram of the GFRP bar-reinforced SSC beam (dimensions in mm) and (B) three-point bending test.

of the SSC beams were prestressed to sufficiently exploit the tensile properties of the GFRP bars. Because excessive prestressing leads to more severe degradation of the GFRP bars³⁷ and because of their creep limit (as described in ACI 440.1R-15),⁴⁶ the prestress level was set to be 20% of the ultimate strength of the GFRP bars.

Three-point bending testing was performed, as shown in Figure 2B. The clear span of each SSC beam was 1000 mm. Linear variable displacement transducers (LVDTs) were deployed at the mid-span and supports of the SSC beams to measure the deflection changes during loading. The test control was based on load control. Before formal loading, the SSC beams were preloaded to 5 kN to eliminate the gaps and then unloaded to zero. Next, the load was applied at a rate of 5 kN/min before concrete

cracking occurred, and the sample was then loaded at 10 kN/min until the SSC beam failed. The crack width of the cracked concrete was measured by a digital crack width meter at 10 kN intervals. In addition, digital image correlation (DIC) was employed to monitor the strain distribution of the SSC beams over an area of 1200 mm × 220 mm. The scatterplot of random black dots used for 2D DIC analysis was sprayed on one side of the beam. A Canon EOS 100d camera with a resolution of 5184 × 3456 pixels was used to capture photographs (per 5 kN) during the loading process. For DIC analysis, a facet size of 19 pixels and a point distance of 16 pixels were selected for DIC postprocessing via the software package GOM Correlate.

2.4 | Prestensioning method

The prestressed SSC beams were prepared by pretensioning. The steps of applying prestressing, binding the FRP bar cages and casting the concrete were as described in the authors' previous work.¹⁷ The target prestress level was 20% of the tensile strength of the GFRP bars. The prestressing was over tensioned by 1.2 times the target stress to minimize the prestressing loss. In addition, a fiber Bragg grating (FBG) with a length of 10 mm was adhered to the surface of the GFRP bar to measure the prestress loss, and the FBG was covered with waterproof glue and an anticollision glue for protection.

2.5 | Environmental and load conditioning

A structure is usually subjected to loads under service conditions. Thus, in this study, for a more realistic match

to the actual conditions, a sustained load was applied to the SSC beams. The process of sustained load application is shown in Figure 3A. A nonprestressed beam was stacked on top of a prestressed beam, and the two beams were supported by two round steel bars between them. Then, a counterframe composed of steel plates and screws was installed in the mid-span of the beams. Next, the jack and load cell were installed, after which the load was applied. After the loading was completed, the nuts on the steel plate were tightened, and the load cells and jacks were removed. To simulate the cracking of nonprestressed beams under service conditions, the sustained load level was 40 kN, which was greater than the cracking load of the nonprestressed control beams and lower than that of the prestressed control beams. The beams were reloaded at monthly intervals, and a deflection measurement was employed to record the deflection of the beams before and after the application of the loads.

After the sustained load was applied, all specimens were exposed to an outdoor real subtropical coastal environment (113.300111 E and 23.132997 N) in Guangzhou, southern China, as shown in Figure 3B. Figure 3C displays the variations in temperature and relative humidity during environmental exposure for the conditioned beams. Guangzhou features an oceanic subtropical environment with abundant rainfall resources and high average annual temperatures. The mean temperature and relative humidity recorded throughout the exposure time were $28.1 \pm 8.77^{\circ}\text{C}$ and $58.6 \pm 19.1\%$, respectively.

2.6 | Scanning electron microscopy (SEM) and alkalinity measurements of the SSC

To further clarify the degradation mechanism of the GFRP bars, the cross-sections of the GFRP bars coated

TABLE 2 Summary of the specimens.

Specimen type	Exposure environment	Exposure time (days)				
		Control	90	180	270	540
Nonprestressed SSC beams	Subtropical coastal environment	1	1	1	1	1
Prestressed SSC beams		1	1	1	1	1
SSC-coated GFRP bars		4	4	4	4	4

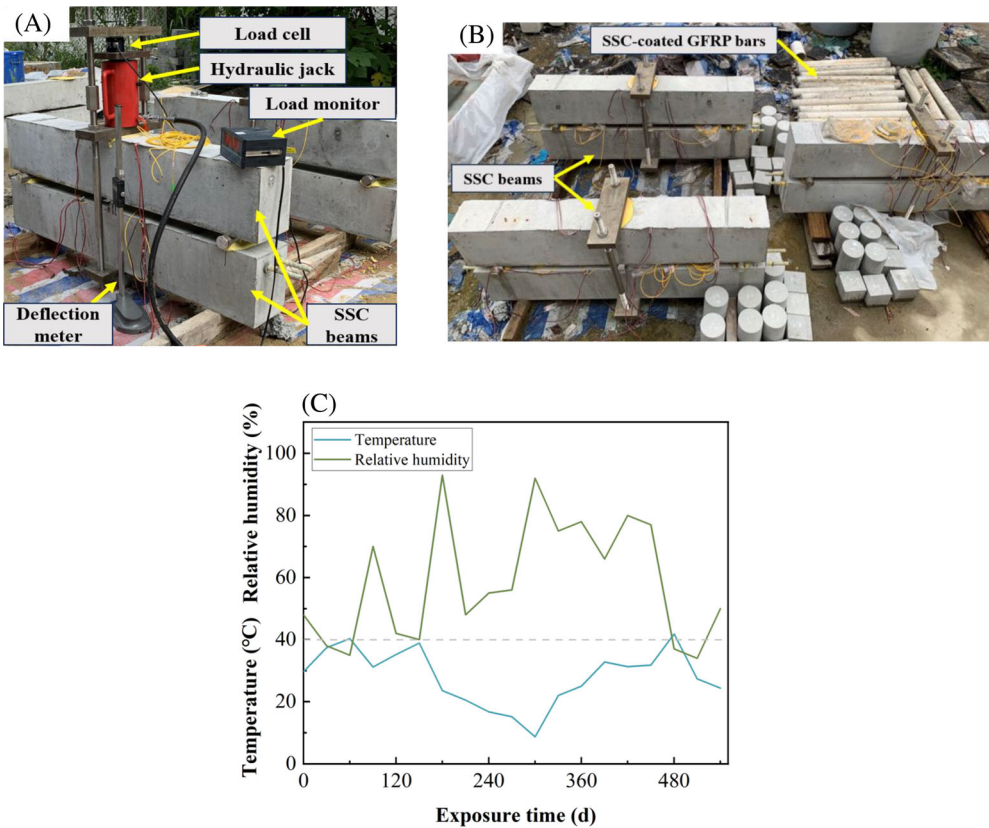


FIGURE 3 Conditions during environmental exposure of the specimens: (A) sustained load application; (B) specimens exposed to a subtropical coastal environment; and (C) temperature and relative humidity during environmental exposure.

with SSC were subjected to SEM. All test samples were sprayed with gold to improve their electrical conductivity. Microscopy images of the cross-sections of the GFRP bars were obtained with an S-3400 N-H microscope.

The method for measuring SSC alkalinity was described previously.³⁷ The SSC at the concrete interface between the SSC and GFRP bars was scraped into a powder, and then the SSC powder was passed through a 0.08 mm sieve. Then, the SSC powder and distilled water were mixed at a ratio of 1:10. After stirring for approximately 30 min, the pH of the suspension was measured with a pH meter.

3 | RESULTS AND DISCUSSION

3.1 | Tensile results for the GFRP bars

A total of 20 GFRP bar specimens (4 controls and 16 conditions) were subjected to tensile tests, and the average ultimate tensile strength and average elastic modulus and their corresponding coefficients of variation (CoVs) for each group of specimens are reported in Table 3. The GFRP bars failed due to glass fiber rupture, as presented in Figure 4A. The evolution of the tensile strength and elastic modulus of the GFRP bars vs. exposure time is plotted in Figure 4B. As shown, the tensile strength of

the conditioned GFRP bars decreased significantly within 90 days of exposure and then continued to decrease gradually. Similarly, compared with those of the control GFRP bars, the tensile strength retention results of the conditioned GFRP bars were 84.25%, 82.56%, 81.32% and 73.22% at 90, 180, 270 and 540 days, respectively. The degradation in the tensile strength of the GFRP bars was primarily induced by the hydroxide ions in the pore solution within the SSC. The hydroxide ions react chemically with glass fibers and vinyl resin,^{47–49} causing deterioration of the interface between the fiber and the resin, which ultimately results in degradation of the tensile strength of the GFRP bars.

Notably, the change in the elastic modulus retention of the GFRP bars was insignificant compared to the change in the tensile strength retention, as shown in Figure 4B. The elastic modulus retention results of the GFRP bars at 90, 180, 270 and 540 days were 97.72%, 89.87%, 88.98% and 93.51%, respectively. This indicated that the alkalinity of the SSC had a minor effect on the elastic modulus of the GFRP bars. Compared with 270 days, the slight increase in the elastic modulus of the GFRP bars at 540 days may have been due to the postcuring effect of the resin. It has been reported that postcuring FRP materials has a positive effect on the enhancement of the mechanical properties of FRP materials.⁵⁰ However, because the tensile strength of

TABLE 3 Results of tensile tests of GFRP bars.

Specimen-duration	Tensile strength (MPa)			Elastic modulus (GPa)		
	Average value	Retention (%)	CoV (%)	Average value	Retention (%)	CoV (%)
GFRP-0 days	1102.11	100.00	4.80	51.72	100.00	3.62
GFRP-90 days	928.57	84.25	1.72	50.54	97.72	4.55
GFRP-180 days	909.88	82.56	1.27	46.48	89.87	3.33
GFRP-270 days	896.20	81.32	3.25	46.02	88.98	5.83
GFRP-540 days	806.98	73.22	2.80	46.91	93.51	7.69

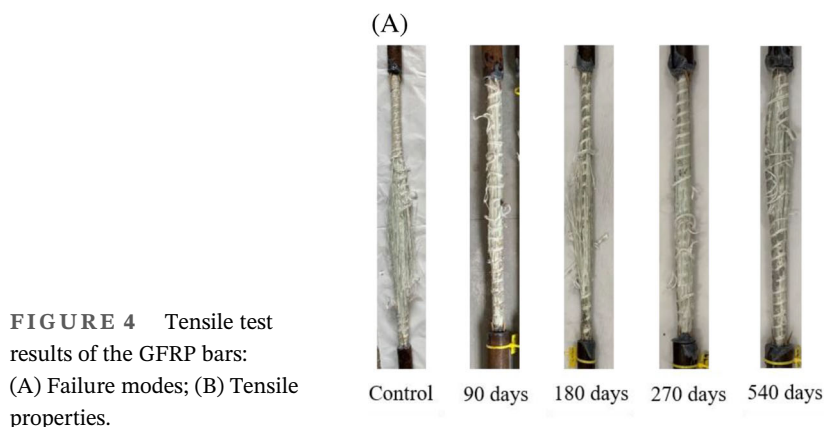


FIGURE 4 Tensile test results of the GFRP bars: (A) Failure modes; (B) Tensile properties.

the GFRP bars was sensitive to the internal defects of the GFRP bars, postcuring of the resin did not significantly improve the tensile strength of the GFRP bars. Therefore, as shown in Figure 4B, the tensile strength of the GFRP bars deteriorated with increasing exposure time, while the elastic modulus improved.

The microscopic morphologies of the control and GFRP bars exposed for 540 days are shown in Figure 5. As shown in Figure 5A, the glass fiber and vinyl matrix of the control GFRP bars were closely attached and maintained good integrity. In contrast, as shown in Figure 5B, the exterior surfaces of the GFRP bars became incomplete after 540 days of exposure, and debonding occurred at the interfaces between the glass fibers and the vinyl matrix. This was because the exterior surface of each GFRP bar was the primary contact area with the pore solution of the SSC; thus, debonding started at the exterior surface of each GFRP bar. The tensile stress was mainly borne by the glass fibers, while the stress could not be equally transferred to some glass fibers due to debonding. Therefore, the remaining glass fibers were subjected to higher stress, which resulted in stress concentrations that led to earlier failure of the GFRP bars.

The change in the pH of SSC with time is presented in Table 4. The alkalinity within the SSC remained constant (~ 12.3) with time. The pH showed little fluctuation, which may have been related to the continuous hydration reaction.^{43,51} However, overall, the GFRP bars were in a stable alkaline environment during environmental exposure.

3.2 | Flexural behavior of GFRP bar-reinforced SSC beams

3.2.1 | Exposure time-dependent prestress level of the GFRP bars

Figure 6 illustrates the evolution of the prestress level of the prestressed SSC beams, where GFRP bars-1 and -2 represent the two prestressed GFRP bars inside the SSC beams. As shown in Figure 6, the process of prestress level evolution consisted of three parts: the jacking stage, curing stage and exposure stage. The prestress loss of the SSC beams during the curing stage was attributed to anchorage relaxation and stress release. After curing for 28 days, the prestress level of the SSC beams was stable at 16%–19%. Most of the prestress loss in the SSC beams during the exposure stage was caused by concrete creep and relaxation of the prestressed GFRP bars due to sustained loading. The average prestress levels of the prestressed SSC beams at 90, 180, and 270 days were 16%, 11% and 9%, respectively. Due to creep relaxation of the prestressed GFRP bars, this may lead to debonding of the FBG from the GFRP bars, and the prestressed SSC beam (P-540d) was measured with an average prestress level at 300 days only.

TABLE 4 SSC alkalinity vs. exposure time.

Exposure time (days)	90	180	270	540
pH of SSC	12.39	12.33	12.39	12.38

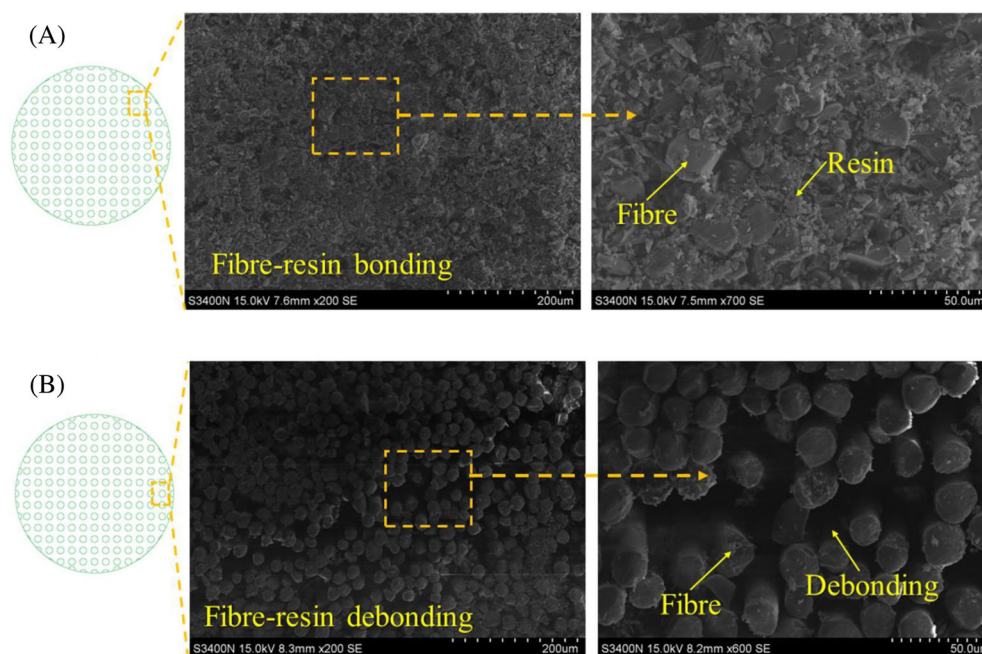
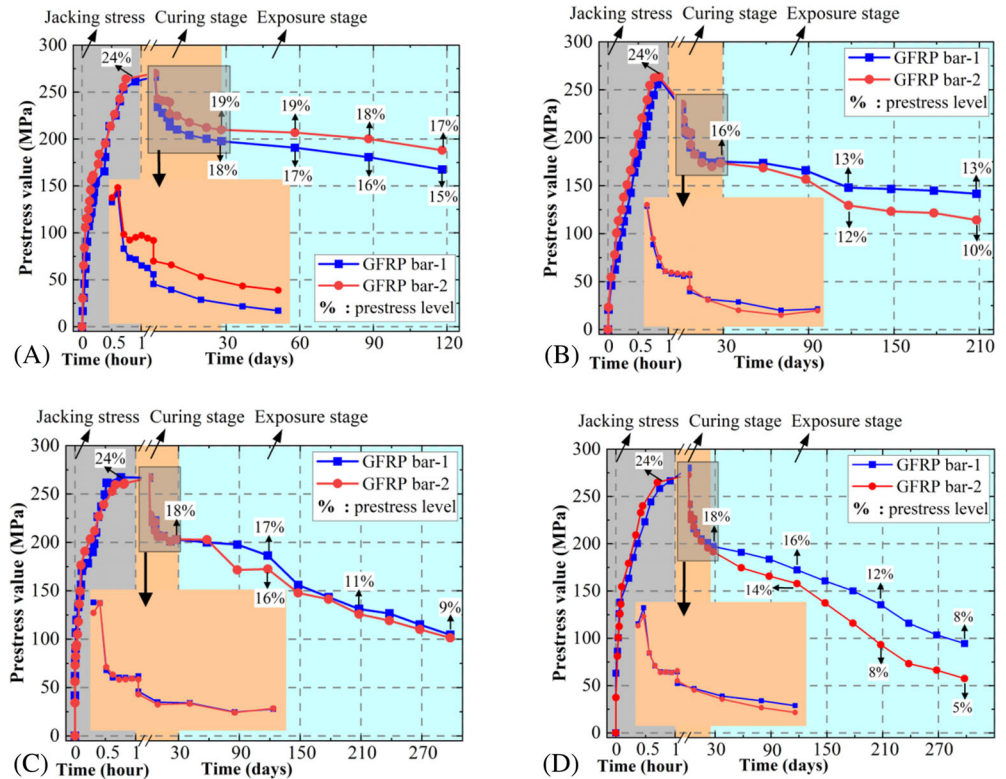


FIGURE 5 Micromorphology of the cross-sections of the GFRP bars: (A) control and (B) exposed for 540 days.

FIGURE 6 Evolution of prestress levels: (A) P-90d; (B) P-180d; (C) P-270d; and (D) P-540d.



3.2.2 | Exposure time-dependent deflections

The conditioned SSC beams exhibited creep behavior; that is, the deflection of the beams increased gradually under the influence of sustained loading. The evolution of the deflection of the conditional SSC beams is shown in Figure 7. The theoretical deflection was calculated according to Equation (1) of the code ACI 440.1R-15,⁴⁶ and the results are plotted in Figure 7. The total theoretical deflection is composed of two parts: the instantaneous deflection $(\Delta_i)_{sus}$ due to the applied sustained load and the deflection $\Delta_{(cp+sh)}$ due to creep and shrinkage. ξ in Equation (1) represents the time factor.

$$\Delta_{(cp+sh)} = 0.6\xi(\Delta_i)_{sus} \quad (1)$$

As shown in Figure 7, both the experimental deflection and theoretical deflection increased rapidly in the earlier days of testing and then plateaued. The ratios of the experimental and theoretical deflections for the prestressed and nonprestressed SSC beams were 1.39 and 0.95, respectively, at 360 days. The theoretical results agreed well with the experimental deflection results of the nonprestressed beams. However, the experimental deflection results of the prestressed beams were greater than the theoretical results due to prestress loss.

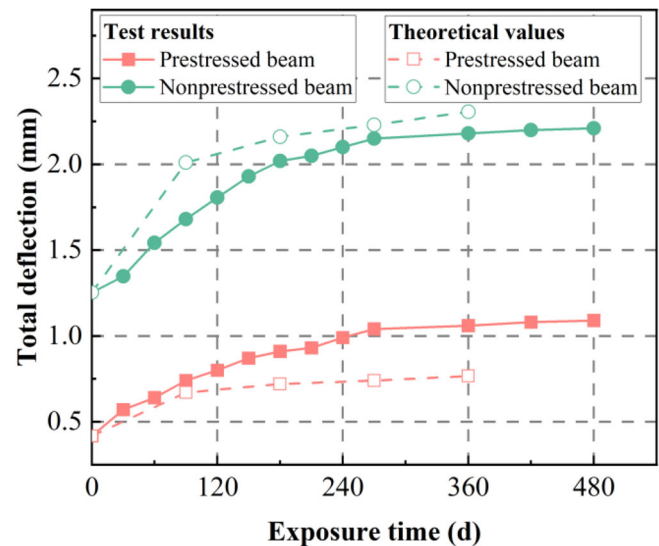


FIGURE 7 Total deflections of the conditioned beam versus time.

3.2.3 | Failure modes of the SSC beams

Figure 8 depicts the failure modes and the final crack distribution at the moment of failure of the SSC beams. The significance of the labels of the SSC beam in Figure 8 is as follows: N/P for nonprestressed/prestressed SSC beams; 90, 180, 270 and 540 days for exposure times; and con for control SSC beams.

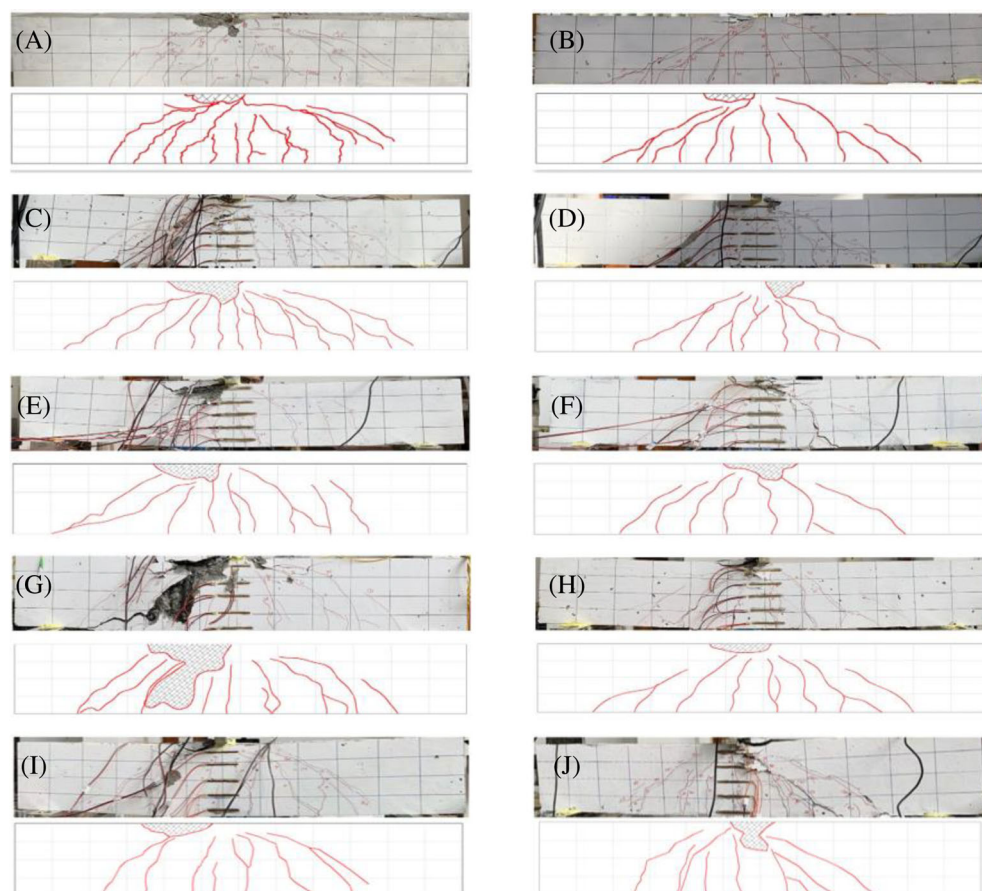


FIGURE 8 Failure modes of the SSC beams: (A) N-con¹⁷; (B) P-con¹⁷; (C) N-90d; (D) P-90d; (E) N-180d; (F) P-180d; (G) N-270d; (H) P-270d; (I) N-540d; and (J) P-540d.

As shown in Figure 8, the control beams (N-con and P-con) both failed when the top concrete was crushed. After exposure, the failure mode of the conditioned SSC beams was concrete crushing, except for that of specimen N-270d (it is possible that the defects inherent in the GFRP stirrups led to a change in the failure mode of N-270d, which is not discussed below for this specimen). This could be attributed to the fact that the SSC beams were designed to be overreinforced, and the failure of the conditioned SSC beams remained concrete crushing after 540 days of exposure. In addition, compared to the corresponding control SSC beams, the 540-day SSC beams had fewer cracks. This may be due to the microcracking of the concrete under the coupled effects of sustained loading and environmental exposure, which deteriorated the bonding between the GFRP bars and the concrete, this is similar to the results previously reported in the literature.²⁹

3.2.4 | Load–deflection curves

Figure 9 depicts the evolutions of the mid-span deflections for all the GFRP bar-reinforced SSC beams during the three-point bending tests. Table 5 displays the

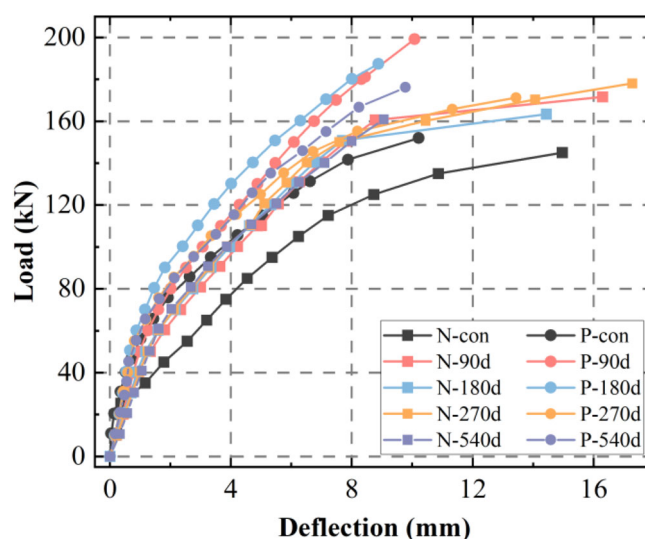


FIGURE 9 The load–mid-span deflection curves of the SSC beams.

characteristic loads and deflections of each SSC beam. As shown in Figure 9, the slope of the load–deflection curve of the SSC beam subjected to environmental effects increased due to the increase in the concrete strength. In particular, the slope of the load–deflection curve of the

TABLE 5 Failure modes and test results of the SSC beams.

Specimens	Cracking load (kN)	Ultimate load (kN)	Maximum deflection (mm)	Failure modes
N-con ¹⁷	25.27	145.01	14.96	Mode I
P-con ¹⁷	46.26	151.93	10.33	Mode I
N-90d	-	171.59	16.29	Mode I
P-90d	50.42	199.22	10.08	Mode I
N-180d	-	163.37	14.44	Mode I
P-180d	60.08	187.44	8.88	Mode I
N-270d	-	178.03	17.28	Mode II
P-270d	54.99	171.25	13.44	Mode I
N-540d	-	160.92	9.06	Mode I
P-540d	-	176.17	9.77	Mode I

Note: Indicates that the SSC beam cracked due to the sustained loading, Mode I denotes that the beam failed by concrete crushing, and Mode II denotes that the beam underwent shear failure.

prestressed SSC beams was steeper than that of the nonprestressed SSC beams for the same exposure time. This may be because the stiffness of the prestressed SSC beams was enhanced by the positive effect of prestressing, while the stiffness of the nonprestressed SSC beams was reduced by cracking due to the sustained loading. In addition, compared to those of the control SSC beams, the ultimate deflections of the nonprestressed SSC beam (N-540d) and prestressed SSC beam (P-540d) decreased by 39.4% and 5.4%, respectively, after exposure for 540 days, as shown in Figure 9. This was due to the creep of the SSC beams under sustained loading, which decreased the bonding between the GFRP bars and the concrete. The reduction in the deflection of the beam was influenced by the loss of bonding between the FRP bars and the concrete, as previously reported in the literature.²⁸

Table 5 presents the changes in the cracking load of the prestressed SSC beams with respect to the exposure time. Similarly, compared with those of the control beam (N-con), the cracking loads of SSC beams P-con, P-90d, P-180d and P-270d increased by 83.1%, 99.5%, 137.7% and 117.6%, respectively. As shown in Figure 6, the prestress levels of SSC beams P-90d, P-180d, and P-270d decreased with time. However, as shown in Table 5, their cracking loads increased. It has been reported that the cracking load of beams is controlled by the tensile strength of the concrete, which depends on the compressive strength of the concrete.^{28,29,52} Therefore, the increased cracking load of the SSC beams in this study can be attributed to the increased compressive strength of the concrete (see Table 1).

As shown in Figure 9, the ultimate flexural capacity of the GFRP bar-reinforced SSC beams was maintained after 540 days of exposure to the subtropical coastal environment. After 540 days, compared with those of the control SSC beams, the ultimate flexural capacities

of the nonprestressed SSC beam (N-540d) and prestressed SSC beam (P-540d) increased by 11.0% and 16.0%, respectively. The increase in the ultimate flexural capacity of the SSC beams could be attributed to the increase in the concrete strength; that is, the failure of the overreinforced SSC beam was determined by the concrete strength.

Wang et al.²⁹ investigated the coupled effect of sustained loading and exposure to a chloride environment on GFRP bar-reinforced SSC beams. A wet-dry cycle of salt solution with 3.5% NaCl was adopted to simulate an environment rich in chloride ions, with a 24-h cycle (12 h of immersion and 12 h of drying). The temperature of the salt solution was 40°C during immersion. Notably, prestressed GFRP bar-reinforced SSC beams exhibited a significant loss of prestress in the chloride environment, after which the prestressed beams cracked due to the sustained loading. However, in the subtropical coastal environment, although there was a gradual loss of prestress in the prestressed GFRP bar-reinforced SSC beams, the cracking load of the prestressed conditioned beams improved by 117.6% after 270 days of exposure to the subtropical coastal environment (as analyzed above).

Furthermore, upon comparing the experimental results from the literature²⁹ and the present work, it was found that the flexural capacity of GFRP bar-reinforced SSC beams in a subtropical coastal environment was greater than that in an accelerated laboratory environment at the same age (270 days). This indicates that the results of the tests in the accelerated laboratory environment were conservative. A major reason for this difference is that the subtropical coastal environment has a milder effect on the beams than does the accelerated laboratory environment. Moreover, because the GFRP bar-reinforced SSC beams in the present work were overreinforced beams, despite the degradation of the GFRP

bars in the beams, the positive effect of the increase in strength of the concrete surpassed the negative effect of the degradation of the GFRP bars in the beams with increasing exposure time. Therefore, the flexural capacity of GFRP bar-reinforced SSC beams in real subtropical coastal environments improved over those of the corresponding control beams.

3.2.5 | Crack width and distribution

Figure 10 depicts the maximum main crack width vs. load for all the SSC beams. The main crack was defined as the initial crack that first appeared in the mid-span of the beam. Due to the inhomogeneity of the concrete material and the dispersion of the bars, there was no distinct regularity in the development of crack width among the SSC beams. The maximum crack width of the conditioned SSC beams was greater than that of the control SSC beams at 90, 180 and 270 days. However, at 540 days, the maximum crack width of the conditioned SSC beam was less than that of the corresponding control SSC beam. This could be explained by the relationship between the development of the crack width in a beam and the deflection of the beam. As shown in Section 3.2.4, the ultimate mid-span deflection of the conditioned SSC beam at 540 days was less than that of the corresponding control SSC beam. In other words, with conditioning, the mid-span deformation of the beam decreased at the same load level, and the crack width decreased accordingly.

The midspan strain contours of the conditioned SSC beams at different load levels obtained based on 19 pixels of facet size and 16 pixels of point distance used for DIC

postprocessing are plotted in Figure 11. The number of mid-span cracks in the prestressed SSC beams under service loading (defined as 60 kN) was less than that in the nonprestressed SSC beams. However, the numbers of mid-span cracks were similar for these beams under the ultimate load. This demonstrated that the development of cracks under service load could be effectively controlled by prestressing.

3.2.6 | Theoretical calculations of the long-term ultimate flexural capacity

The balanced reinforcement ratio of the exposed beams was calculated by Equation (2),⁵³ and the results are listed in Table 6. As shown in Table 6, under environmental conditions, the real reinforcement ratio of the beams was less than 1.5 times the balanced reinforcement ratio due to the decrease in the tensile strength of the GFRP bars. Therefore, using tensile strength of the exposed GFRP bars used to calculate the theoretical flexural capacity of the exposed beams should be calculated by Equation (4) in accordance with GB 50608.⁵³ Specimen N-270d was not subjected to the following calculations because it underwent shear failure.

$$\rho_{fb} = \frac{\alpha_1 f_c}{f_{fd}} \left(\frac{\beta_1 \varepsilon_{cu}}{\varepsilon_{cu} + f_{fd}/E_f} \right) \quad (2)$$

$$f_{fd} = \frac{f_{fu}}{\gamma_f \gamma_e} \quad (3)$$

$$M_u \leq f_{fd} A_f (h_{of} - x/2) \quad (4)$$

$$x = \left(\frac{0.14}{1 + 400(f_{fd}/E_f)} + \frac{\rho_f f_{fd}}{f_c} \right) h_{of} \quad (5)$$

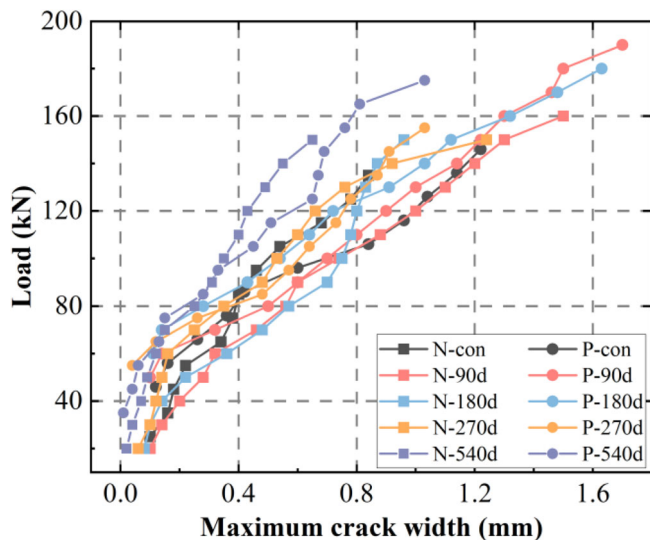


FIGURE 10 The load–maximum crack width curves of the SSC beams.

where f_c is adopted for the SSC compressive strength at different exposure times to better reflect the flexural capacity of the beams, as shown in Table 1; f_{fu} is adopted for the tensile strength of the GFRP bars at each exposure time in Section 3.1; ρ_f is the real reinforcement ratio; ρ_{fb} is the balanced reinforcement ratio; x represents the height of the compression zone of the beam; h_{of} represents the effective height of the beam; γ_e is the environmental reduction factor, which is 1.40 for GFRP bars in a subtropical coastal environment according to code GB 50608⁵³; f_{fd} is the designed tensile strength of the GFRP bars; and α_1 , β_1 , γ_f and others can refer to code GB 50608.⁵³

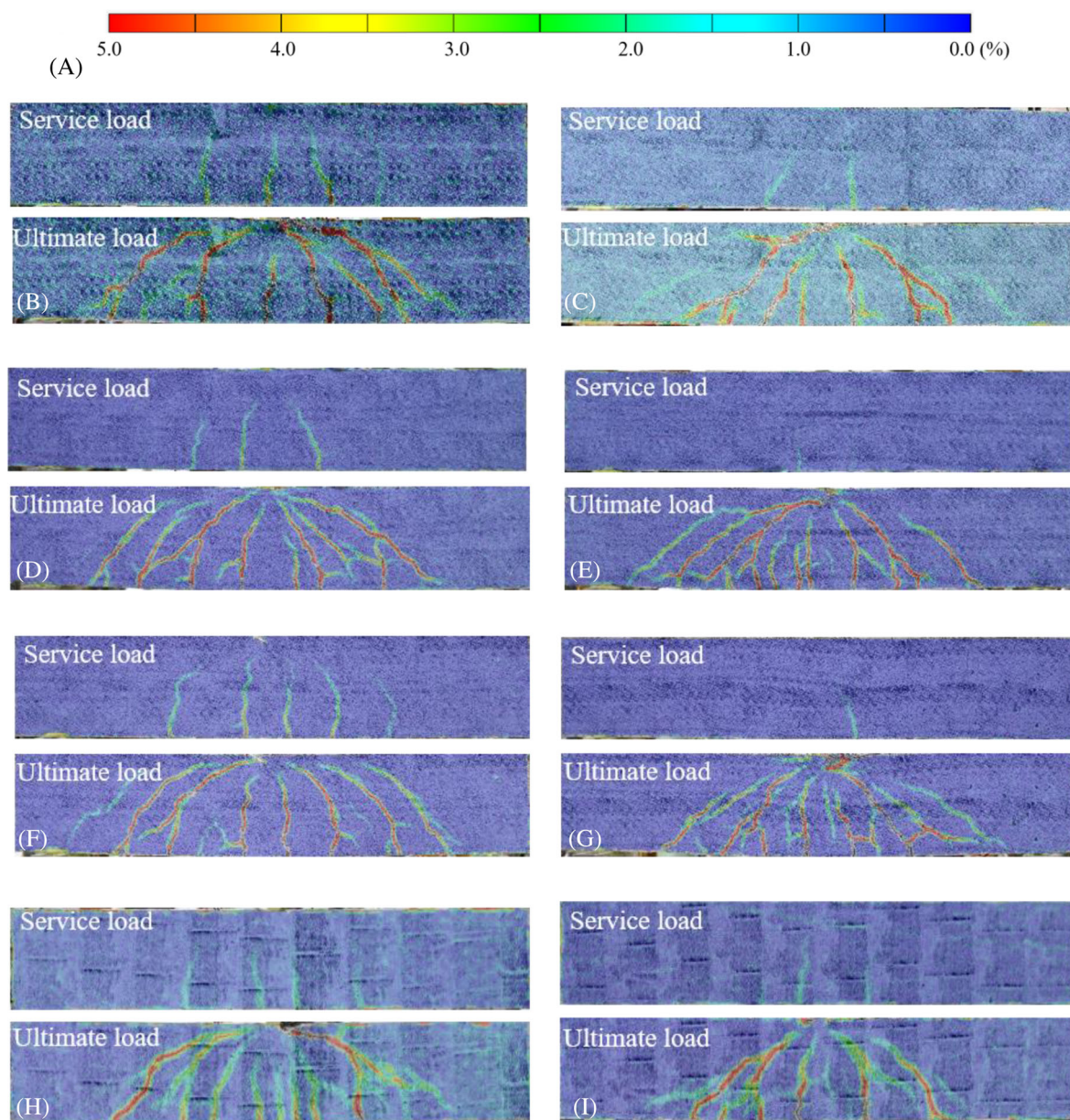


FIGURE 11 Strain contours of the SSC beams under different loads: (A) strain intervals; (B) N-90d; (C) P-90d; (D) N-180d; (E) P-180d; (F) N-270d; (G) P-270d; (H) N-540d; and (I) P-540d.

TABLE 6 Balanced reinforcement ratio of the GFRP bar-reinforced SSC beams.

Exposure time (days)	f_c (MPa)	f_{fu} (MPa)	ρ_f (%)	$1.5 \rho_{fb}$ (%)
0	47.74	1102.11	2.4773	1.9396
90	54.19	928.57	2.4773	2.9309
180	58.18	909.88	2.4773	3.0589
270	59.56	896.2	2.4773	3.1921
540	64.01	806.98	2.4773	4.1865

A comparison of the theoretical and experimental flexural capacities of the exposed beams is shown in Figure 12. As shown, the theoretical results are higher than the experimental results, and the average ratio of

the theoretical to experimental flexural capacities was 1.05 ± 0.06 . This demonstrated that the theoretical results provided nonconservative predictions of the ultimate flexural capacity of the exposed beams. A potential

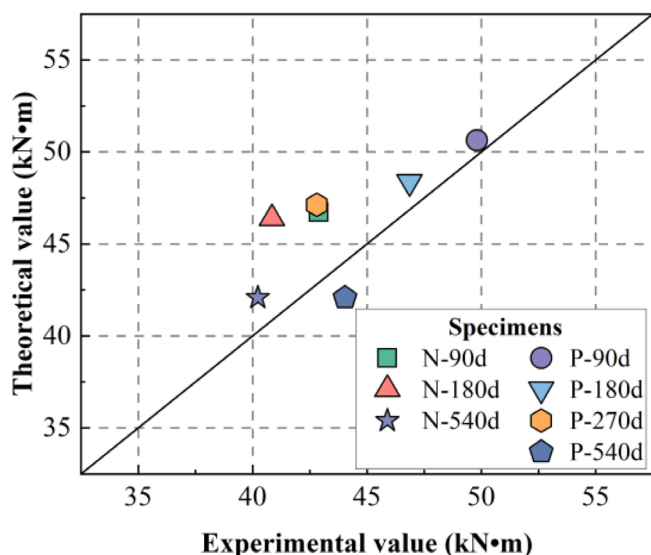


FIGURE 12 Comparison between the theoretical and experimental flexural capacities of the exposed beams.

reason for this finding is that the exposed beams in this paper were subjected to a coupled effect of sustained loading and environmental exposure, which implied that the exposed beams experienced more severe conditions due to this coupling. In the theoretical capacity calculations of the exposed beams described above, only the real tensile strength degradation of the GFRP bars and the increase in the concrete strength were considered, and the negative effect of the sustained loading was not accounted for in the calculations. Notably, concrete structures in real service environments are commonly subjected to loading. Therefore, when calculating the long-term flexural capacity of exposed beams, it is important to consider not only the evolution of the strength of the GFRP bars and concrete but also the influence of external loading conditions in the calculation. However, more data are needed to determine the factors that influence the external loading conditions on the calculation of the theoretical capacity of exposed beams.

4 | CONCLUSIONS

The results of the experimental study reported in this paper highlight the potential of GFRP bar-reinforced SSC beams for coastal engineering applications. The tensile properties of the GFRP bars embedded in SSC and the flexural capabilities of the SSC beams reinforced with GFRP bars were investigated after exposure to a subtropical coastal environment. The following conclusions were drawn from the above findings:

The SSC beams reinforced with GFRP bars performed well in the subtropical coastal environment. The chloride

ions contained in the SSC did not have an obvious detrimental effect on the flexural behavior of the conditioned SSC beams.

SEM observations showed that the GFRP bars embedded in SSC underwent fiber–resin debonding at the outer edges. The tensile strength of GFRP bars decreased by 27% after 540 days of exposure to a subtropical coastal environment.

Compared to that of the nonprestressed SSC beams in the reference group, the cracking load of the prestressed SSC beams conditioned for 270 days was significantly enhanced by approximately 117.6%. This reflected the positive effects of prestressing and concrete strength enhancement (due to continued hydration reactions) on the cracking resistance of the conditioned SSC beams.

The coupled effects of sustained loading and exposure to the subtropical coastal environment had little influence on the failure mechanism and residual strength of the SSC beams reinforced with GFRP bars. After exposure for 540 days, the flexural strength of the nonprestressed SSC beams and prestressed SSC beams improved by 11.0% and 16.0%, respectively, compared to that of their control counterparts.

The creep of the conditioned SSC beams induced by a sustained load increased rapidly before 150 days but the increase in creep gradually slowed down with increasing exposure.

The flexural capacity of the exposed beams was unconservatively predicted by integrating the effect of the tensile strength degradation of the GFRP bars. The calculation of the theoretical flexural capacity of the exposed beams was related to the external loading conditions, and the relevant functional formulas needed to be established.

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DATA AVAILABILITY STATEMENT

All the data, models, and/or codes are available from the corresponding author.

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